

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T53 · Statistics

Questions

Student book · questions with writing space

■ 41 pages

FAMILY MAP

Statistics Toolkit

F01	Estimate grouped-data measures	14 Q 44 marks
F02	Calculate and reverse discrete averages	16 Q 51 marks
F03	Measure and compare spread	6 Q 14 marks
F04	Present and interpret statistical diagrams	3 Q 8 marks

1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

(a) Write down the modal class.

.....
(1)

(b) Work out an estimate for the mean distance.

..... km
(4)

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

Question 12

4MA1-2H Q1

MODULAR UNIT 2

3 marks

- 1 Here are eight numbers written in order of size

 $h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

 $h = \dots\dots\dots$ $j = \dots\dots\dots$ $k = \dots\dots\dots$

4MA1-2H Q12

3 marks

- Work out his mean number of steps for the 7 days.

.....

Question 21

4MA1-2H Q1 (a)

MODULAR UNIT 2

1 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (a) Write down the modal class.

(1)

Question 22

4MA1-2H Q1 (b)

MODULAR UNIT 2

3 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (b) Work out an estimate for the total length, in minutes, of these telephone calls.

..... minutes

(3)

Question 27

4MA1-2H Q8 whole

MODULAR UNIT 2

3 marks

- 8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

$k =$

Question 28

4MA1-2H Q13 whole

MODULAR UNIT 2

2 marks

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

.....

Question 02

4MA1-1H Q2

MODULAR UNIT 2

3 marks

- 2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leq 30$	5
$30 < M \leq 60$	6
$60 < M \leq 90$	8
$90 < M \leq 120$	9
$120 < M \leq 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

..... minutes

Question 10

4MA1-1H Q10

MODULAR UNIT 2

2 marks

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

.....

Question 23

4MA1-2H Q10

MODULAR UNIT 2

4 marks

10 Team A and Team B take part in a quiz league.

After 11 rounds, Team A has a mean score per round of 17

After 9 rounds, Team B has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team A scored more points than Team B.

How many more?

Question 01

4MA1-1H Q1 M01

MODULAR UNIT 2

1 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (a) Write down the modal class.

(1)

Question 02

4MA1-1H Q1 M02

MODULAR UNIT 2

4 marks

1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

(b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

4MA1-2H Q8 M01

3 marks

- grams

Question 24

4MA1-2H Q10 M01

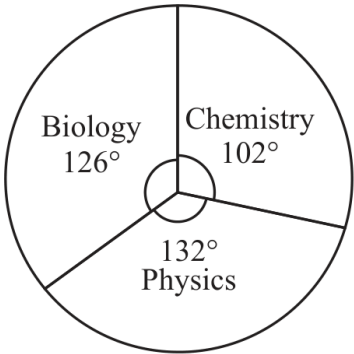
MODULAR UNIT 2

5 marks

- 10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.

The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10

There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

Question 24 (continued)

4MA1-2H Q10 M01

MODULAR UNIT 2

5 marks

.....

Lined area for student response.

Question 20

4MA1-2H Q1 Q1

MODULAR UNIT 2

3 marks

- 1 Here are some integers where $a < b < c < d$

$a \qquad b \qquad c \qquad d \qquad d \qquad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

$d = \dots\dots\dots$

Question 25

4MA1-2H Q12 Q12

MODULAR UNIT 2

3 marks

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

$W =$

Question 04

4MA1-1H Q5 Q5(a)

MODULAR UNIT 2

3 marks

5 Jenny has six cards.

Each card has a whole number written on it so that

the smallest number is 5

the largest number is 24

the median of the six numbers is 14

the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

(a) For the remaining four cards, write on each dotted line a number that could be on the card.

Question 05

4MA1-1H Q5 Q5(b)

MODULAR UNIT 2

3 marks

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

(b) Work out the number of points the team scored in its 6th game.

(3)

Question 23

4MA1-2H Q9

MODULAR UNIT 2

4 marks

- 9 The frequency table gives information about the number of points scored by a player.

Number of points	Frequency
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

4MA1-2H Q15

3 marks

- km

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

(a) Write down the modal class.

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

(a) Write down the modal class.

(b) Work out an estimate for the mean weight of the 40 babies.

..... kg
(4)

..... kilometres per hour
(4)

For these five numbers

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.

..... grams

Here are the results.

Find the interquartile range of these times.

..... minutes

Work out the least mark that Jethro needs to get for the last test.

5 8 9 11 13 13 14 15 16 17 20

Find the interquartile range of the numbers of goals.
Show your working clearly.

..... euros
(4)

4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

- 2 The table shows information about the lengths of time, in minutes, 120 customers spent in a supermarket.

Length of time (L minutes)	Frequency
$20 < L \leq 30$	6
$30 < L \leq 40$	26
$40 < L \leq 50$	31
$50 < L \leq 60$	40
$60 < L \leq 70$	17

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean length of time spent by the 120 customers in the supermarket.

.....minutes
(4)

Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

Work out an estimate for the mean length of time the students spent dancing.

..... minutes

- 15** Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

(Total for Question 15 is 3 marks)

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

e	f	7	g	12	17
-----	-----	---	-----	----	----

The integers have

- a range of 15
a median of 8.5
a mean of 9

$e =$

$$f = \dots\dots\dots$$

$g =$

(Total for Question 4 is 3 marks)

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)